



# Lake Baikal

*Earth's deepest freshwater lake, containing more water than any other and home to many endemic species*

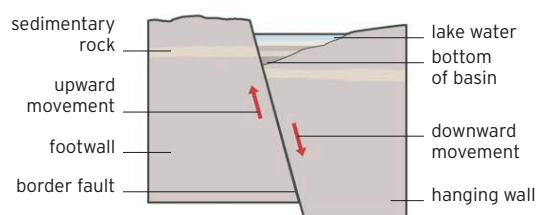
Lake Baikal is a Russian lake situated to the south of the Central Siberian Plateau, near Mongolia. The largest freshwater lake on Earth by volume, it contains around 20 percent of the world's total surface freshwater—similar to that of North America's Great Lakes combined (see pp.50–51). At 395 miles (636 km) long and up to 49 miles (79 km) wide, Baikal has the largest surface area of any freshwater lake in Asia. Its waters have a maximum depth of 5,370 ft (1,637 m), making it the deepest body of freshwater on Earth.

## Ancient and dynamic

Also the oldest lake in the world, Lake Baikal formed in a deep rift valley—where the Earth's crust is pulling apart—more than 25 million years ago. The southwest-to-northeast fault along which the lake runs is still active, and the lake is widening by more than  $\frac{3}{4}$  in (2 cm) each year. Lake Baikal's water is renowned for its purity and clarity. Well oxygenated and well mixed throughout its depths, it supports a wide variety of plants and animals, and more than 80 percent of these species are thought to be found only in Lake Baikal.

## HALF-GRABEN BASINS

The type of basin in which Lake Baikal sits is known as a half-graben. It was formed through the vertical movement of the Earth's crust along a fault, with the fault bordering just one side. In a full graben, a basin is formed by a depressed block of land bordered on both sides by faults.



## ▷ LATE FREEZE

Lake Baikal is completely covered with ice for 4-5 months a year. The huge volume of water in its basin means the surface freezes later in the Siberian winter than the region's other lakes. Peschanaya Bay and Baikal's other bays are the first parts of the lake to freeze over.



## ► THREE BASINS

Lake Baikal is divided into three basins: North, Central, and South. The deepest parts of the lake are in the Central and South Basins. Both the lake's main source and its single outlet are in the South Basin.

**Single outlet**  
Most lakes have a single outflow into a river system. Lake Baikal drains into the Angara River, a tributary of the Yenisei.



## △ FRESHWATER SEAL

The endemic Baikal seal is the only exclusively freshwater seal species in the world. One of the smallest seals, it grows up to 4½ ft (1.4 m) in length.

## Protected area

Several areas around the lake are protected. The Baikal Nature Reserve by the southeastern shore is part of the World Network of Biosphere Reserves.